

Year Up United

Data Analyst Training Academy:

Advanced Reports with Power BI

Week 11 Lab Workbook

This courseware is to be distributed to accompany instructor led training.

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Module 1

Advanced DAX Concepts

Exercise 1.A Northwind – Resolving Granularity Issues

Introduction

Each table in a Power BI Model should either be a **data** (fact) table or a **lookup** (dimension) table. Data tables should never be related to each other. Instead, data tables should be related to lookup tables, and it is only through lookup tables that data tables should relate – representing the dimensions that they have in common.

In this exercise we are going to consider a challenge with our current Northwind Model. We have two related data tables, the Orders table, and the Sales (originally Order Details) table, and these both contain measures at different grains.

The **Orders** table has the following measure: **Order Freight Amount**

The **Sales** table has many measures and is more granular than the Orders table. If we create a report that puts the `Sum(Sales[Sales Amount])` side by side with the `Sum(Orders[Order Freight Amount])`, we will find that the Orders record repeats for each product in the order, which incorrectly inflates the sum of Order Freight Amount.

We want to move the **Order Freight Amount** from the **Orders** table to the **Sales** table.

Our goal is that if we total these two side by side, that totals related to **Freight** can be used to create insight into how individual products influence freight costs at the order level.

There's a business rule that the **Freight Amount** belongs to the order as a whole. Individual products don't directly drive the cost of **Freight** because we're putting multiple products in a box and shipping an order as a unit. So, how might we divide up the freight cost across products in the box?

A compromise might be to split the cost of shipping across each Sales line based on the percentage of the total quantity of that order that the sales line represents. This isn't a perfect solution, but it may provide some insight into our shipping costs so that we can start to see trends around what items are costing to ship.

1. Open the Northwind Modeling project that you have been building throughout the course: **NorthwindModeling_Step_12.pbix**
2. The first objective is to understand the total number of items being shipped in an order. Using the **Table view**, create a calculated column on the **Orders** table named

[Order Quantity]. Set it equal to the sum of the Sales[Sales Quantity] column for each of the rows in the **Sales** table that relate to this row in the Orders table.

* Hint: You will need to use the `RELATEDTABLE()` and `SUMX()` functions.

3. Hide both the [Order Quantity] and [Order Freight Amount] columns in the Orders table, so that they will not be directly used in reports. Both are not the correct granularity to be used in this model.
4. Add a calculated column to the Sales table named [Percent of Order Quantity], and set it equal to the percentage of the order quantity that this specific line item represents. Pull the order quantity from the column you created in the Order table.
* Hint: Divide the Sales[Sales Quantity] by `RELATED(Order[Order Quantity])`
5. Format [Percent of Order Quantity] as a percentage and hide it from the report view. This is a supporting calculation and should not be used directly in reports.
6. Add a calculated column to the Sales table named [Sales Freight Amount]. Set it equal to the Order Freight Amount multiplied by the Percent of Order Quantity.
* No hint on this one, but if you're stuck you might reflect on previous hints!
7. Add a calculated column to the Sales table named [Sales Amount w/ Freight]. Set it equal to the Sales Amount plus the Sales Freight Amount.
8. Add a measure to the Sales table named Sales Freight Percent. Use the following expression:
$$\text{Sales Freight Percent} = \frac{\text{Sum}(\text{Sales}[\text{Sales Freight Amount}])}{\text{Sum}(\text{Sales}[\text{Sales Amount w/ Freight}])}$$
9. Set the format of [Sales Freight Percent] to display as a percentage.
10. Switch to the **Report view**. Create a new report page, and name it **Freight Reports**.
11. Using the values that you added to the data model above, create a report that identifies the:
* **Product Category** with the highest Freight Percent?
* **Product** with the highest Freight Percent?

- * **Country** with the highest Freight Percent?
- * The **Category** with the highest freight percent within the United States?
- * What **Country** has the highest freight percent when shipping products in the **Beverages Category**?

12. Check your answers. Your report findings should show:

- * Seafood
- * Konbu
- * Argentina
- * Seafood
- * Italy

Note: These numbers are prone to error because of the way that this takes an order level value and spreads it out among its children.

We could have selected a few different ways to have spread the freight costs out. If this was an important metric, we might take some time to experiment with what effect various calculation methods may have:

- * We could divide the freight amount equally among line items. Four lines would each get 25% of the cost of the freight, regardless of **Quantity** per line.
- * We could utilize the percentage that a single line represents of the **Sales Amount** for the total order, instead of the percentage of the quantity.
- * We could have investigated other business details around what causes a product's freight to increase, such as weight and size.

Your choices to handle a calculation one way versus another will often be influenced by the purpose of the report you are creating. Perhaps we are considering offering Free Shipping promotions, and we want to understand what the cost of such a promotion would be, and raise prices to offset the cost of the promotion for product groups that are most affected. Take a moment to reflect, what other business decisions might this information be used for?

13. Save your work as **NorthwindModeling_Step_13.pbix**

Exercise 1.B Northwind – Row Level Security

Setting RLS Roles:

1. Continue working with the Northwind Modeling project that we have been building throughout the course.
2. On the **Modeling** tab of the ribbon in the Report view, select **Manage Roles**.
3. Click **New** to create a new role, and name it **US Only**.
4. Click on the Customers table, then add a condition to require Customer Country to equal USA.
5. Click **Save**, then click Close return to the Report view.
6. Navigate to the **Freight Reports** tab in the Reports view. Notice this shows data for many countries.
7. On the **Modeling** tab of the ribbon select **View as**.
8. Check **US ONLY** and click **OK**. Observe that the report now limits data to a single country.
9. Navigate through the other reports and notice that every one of them now returns a smaller subset of data.
10. Repeat the steps above to create another security role for **Americas**, and set a condition on Customer Country using “in” to include Argentina, Brazil, Canada, Mexico, USA, and Venezuela. Then use View As to preview the Americas role and ensure it is working as expected.
11. Save your work as **NorthwindModeling_Step_14.pbix**

Module 2

Date and Time Intelligence

Exercise 2.A Northwind – Adding a Date Table & Hierarchy

Computing a Date Table

In this exercise, you will create a **Date** table for your model using DAX.

1. Continue working with the Northwind Modeling project that you have been building throughout the course.
2. In the ribbon navigate to **File > Options and Settings > Options > Current File > Data Load**
3. Unselect the **Auto date/time** option under **Time intelligence**.
4. Click **OK** to close the **Options** dialog.
5. In the **Report** view, select the **Modeling** tab within the **Ribbon**.
6. Select **New table** in the ribbon.
7. Enter the following query into the expression bar. Save yourself some time by cut-and-pasting it from the text file in the course material, **DAX_DateTableExpression.txt**

```

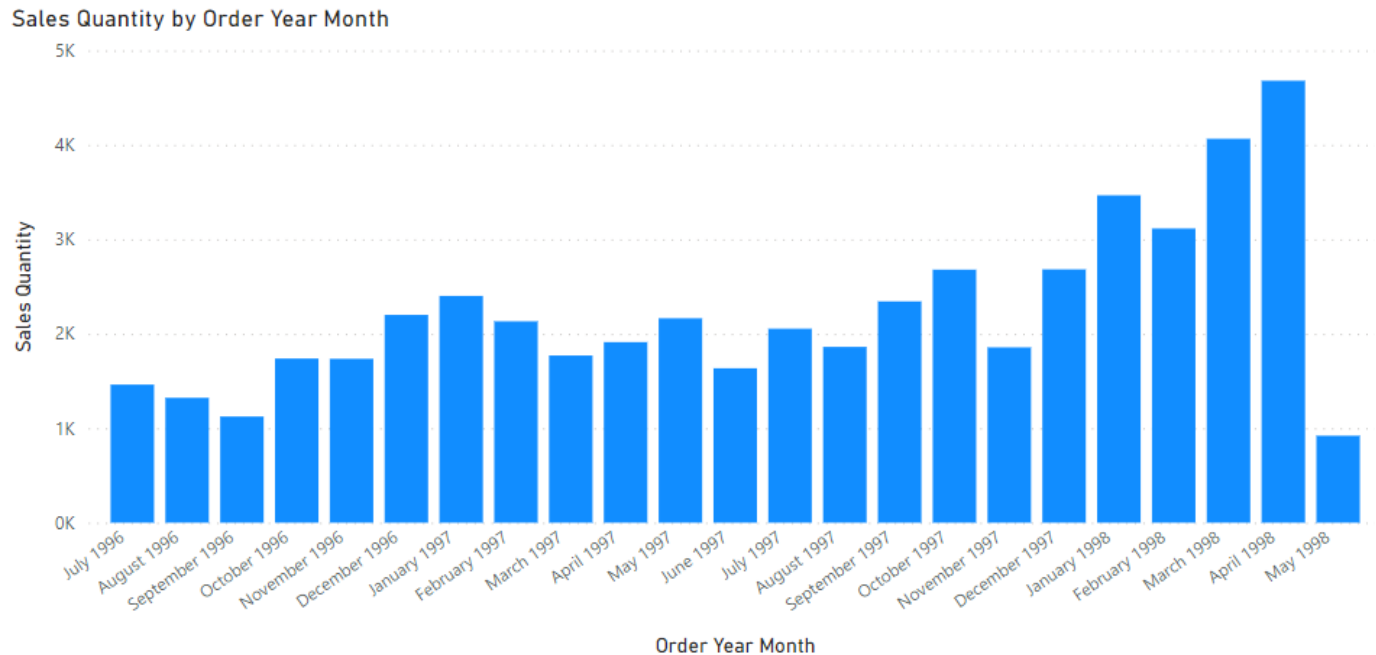
1 Order Date =
2 VAR MinYear = YEAR( MIN('Order'[Order Date]))
3 VAR MaxYear = YEAR( MAX('Order'[Order Date]))
4 RETURN
5 ADDCOLUMNS (
6     FILTER (
7         CALENDARAUTO ( ),
8         YEAR ( [Date] ) >= MinYear &&
9         YEAR ( [Date] ) <= MaxYear
10    ),
11    "Order Year", YEAR ( [Date] ),
12    "Order Quarter Number", INT ( FORMAT ( [Date] , "q" ) ),
13    "Order Quarter", "Q" & INT ( FORMAT ( [Date] , "q" ) ),
14    "Order Month Number", MONTH ( [Date] ),
15    "Order Month", FORMAT ( [Date], "mmm" ),
16    "Order Week Day Number", WEEKDAY ( [Date] ),
17    "Order Week Day", FORMAT ( [Date], "ddd" ),
18    "Order Year Month Number", YEAR ( [Date] ) * 100 + MONTH ( [Date] ),
19    "Order Year Month", FORMAT ( [Date], "mmm" ) & " " & YEAR ( [Date] ),
20    "Order Year Quarter Number", Year ( [Date] ) * 100 + INT ( FORMAT ( [Date], "q" ) ),
21    "Order Year Quarter", "Q" & FORMAT ( [Date], "q" ) & "-" & YEAR ([Date] )
22 )

```

8. Click the check mark save the query.
9. Navigate to the **Data** view, and confirm you have data in the new table.
10. Rename the **Date** column to **Order Date**.
11. Change the Order Date column type from **Date/Time** to **Date** to remove the time component of the value.
12. In the **Orders** table, change the Order Date, Order Required Date, and Order Shipped Date columns to the **Date** data type to remove the time component of the value.
13. In the **Model** view, in the **Home** tab in the ribbon, click **Manage Relationships**.
14. Click **New**.
15. In the **New Relationship** dialog, create a relationship between the **Order** table and **Order Date** table.
 - * Be sure to select the Order Date column from the Order table, and the Order Date column from the Order Date table.

Cardinality:	Many to one (*:1)
Cross filter direction:	Single
Make this relationship active:	checked
16. Back in the Data view, with the **Order Date** table selected, click **Mark as date table** in the **Table tools** tab of the ribbon. If prompted, use the **Order Date** column as the date.
17. In the Model view, configure the **Sort by column** for the following columns:
 - * **Order Year Month** column: sort by the Order Year Month Number column
 - * **Order Month** column: sort by the Order Month Number column
 - * Apply a similar sort of pattern to the **Order Year Quarter**, **Order Week Day**, and **Order Quarter** columns
18. Create a new report page named **Dates 1**.

19. Add a column chart that looks like this:



20. Save your work as **NorthwindModeling_Step_15.pbix**

Adding a Custom Date Hierarchy

1. Continue working with the Northwind Modeling project.
2. Using the **Model** view, create a **Hierarchy** of the Order Date column of the Order Date table.
3. Name the hierarchy **Order Date Hierarchy** and add Order Year, Order Quarter, and Order Month columns.
4. Move the Order Date column below the others in the hierarchy and then select **Apply Level Changes**. When you're done the hierarchy should look like this:

Hierarchy

Select a column to add level... ▼

✦ Order Year (Order Year) ×

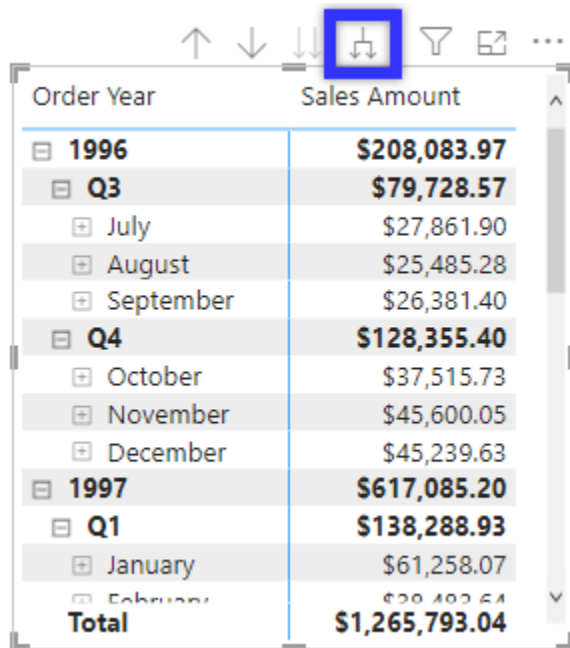
✦ Order Quarter (Order Quarter) ×

✦ Order Month (Order Month) ×

✦ Order Date (Order Date) ×

Apply Level Changes

5. Create a new report page named **Dates 2**.
6. On the new report page, add a matrix visualization with the following fields:
 - * Rows: 'Order Date'[Order Date Hierarchy]
 - * Values: Sales[Sales Amount]
7. With the matrix selected, click the "upside down pitchfork" twice to expand down two levels within the hierarchy. Your visualization should then look like this:



Order Year	Sales Amount
1996	\$208,083.97
Q3	\$79,728.57
July	\$27,861.90
August	\$25,485.28
September	\$26,381.40
Q4	\$128,355.40
October	\$37,515.73
November	\$45,600.05
December	\$45,239.63
1997	\$617,085.20
Q1	\$138,288.93
January	\$61,258.07
February	\$28,482.64
Total	\$1,265,793.04

8. Save your work as **NorthwindModeling_Step_16.pbix**

Exercise 2.B Northwind – Running Totals & Parallel Periods

Compute Running Totals

1. In your Northwind project, add the following measures to the Sales table:

```
Sales Amount YTD = CALCULATE(SUMX(sales, Sales[Sales Amount]), DATESYTD('Order Date'[Order Date]))
```

```
Sales Amount QTD = CALCULATE(SUMX(sales, Sales[Sales Amount]), DATESQTD('Order Date'[Order Date]))
```

2. Set the data type for the new measures to **currency**.
3. Experiment with adding your new measures to the **Dates 2** report page.

Order Year	Sales Amount	Sales Amount QTD	Sales Amount YTD
[-] November	\$43,353.81	\$110,283.053	\$343,080.773
[-] December	\$71,398.43	\$181,681.4635	\$617,085.2035
[-] 1998	\$440,623.87		\$440,623.866
[-] Q1	\$298,491.55	\$298,491.553	\$298,491.553
[-] January	\$94,222.11	\$94,222.1105	\$94,222.1105
[-] February	\$99,415.29	\$193,637.398	\$193,637.398
[-] March	\$104,854.16	\$298,491.553	\$298,491.553
[-] Q2	\$142,132.31	\$142,132.313	\$440,623.866
[-] April	\$123,798.68	\$123,798.6825	\$422,290.2355
[-] May	\$18,333.63	\$142,132.313	\$440,623.866
[-] June		\$142,132.313	\$440,623.866
[-] Q3			\$440,623.866
[-] July			\$440,623.866
[-] August			\$440,623.866
[-] September			\$440,623.866
[-] Q4			\$440,623.866
[-] October			\$440,623.866
[-] November			\$440,623.866
[-] December			\$440,623.866
Total	\$1,265,793.04		\$440,623.866

4. Save your work as **NorthwindModeling_Step_17.pbix**

Compare Parallel Periods

1. Add the following measures to the **Sales** table:

```
Prior Year Sales Amount YTD = Calculate(SUM(Sales[Sales  
Amount]), DATESYTD(SAMEPERIODLASTYEAR('Order Date'[Order  
Date])))
```

```
Change in YTD from Prior Year = Sales[Sales Amount YTD] /  
Sales[Previous Sales Amount YTD]
```

2. Set **Prior Year Sales Amount YTD** to use the **currency** data type.
3. Set **Change in YTD from Prior Year** to be formatted as a **percentage**.
4. Add these fields to your matrix on report **Dates 2**.
5. Spend a few moments exploring your data.
6. Save your work as **NorthwindModeling_Step_18.pbix**

Module 3

Practical Modeling Techniques

Exercise 3.A Working with Nutrition Data

Description

In this workshop, you will walk through end-to-end creation of a Power BI report, from ingesting, shaping, and modeling the data, to creating visualizations and report interactions, and finally publishing the report and creating an online dashboard.

Let's import some data!

1. Open a new blank Power BI project. Save as **nutrition data.pbix**
2. From the USDA site [Food and Nutrient Database for Dietary Studies \(FNDDS\)](https://www.ars.usda.gov/ARSUserFiles/80400530/apps/2019-2020%20FNDDS%20At%20A%20Glance%20-%20FNDDS%20Nutrient%20Values.xlsx), import both of the following data sets using the Get Data > **Web** option by pasting in the link text below (you will need to separately import each file one at a time):

* [FNDDS Nutrient Values.xlsx](https://www.ars.usda.gov/ARSUserFiles/80400530/apps/2019-2020%20FNDDS%20At%20A%20Glance%20-%20FNDDS%20Nutrient%20Values.xlsx)

<https://www.ars.usda.gov/ARSUserFiles/80400530/apps/2019-2020%20FNDDS%20At%20A%20Glance%20-%20FNDDS%20Nutrient%20Values.xlsx>

Import the detected table **FNDDS_Nutrient_Values**:

The screenshot shows the Power BI Navigator window. On the left, under 'Display Options', the 'FNDDS_Nutrient_Values' table is selected with a blue border and a green checkmark. On the right, a preview of the table is shown, titled 'FNDDS_Nutrient_Values' with a subtitle 'Preview downloaded on Monday, July 14, 2025'. The table has four columns: 'Food code', 'Main food description', 'WWEIA Category number', and 'WWEIA Cat'.

Food code	Main food description	WWEIA Category number	WWEIA Cat
11000000	Milk, human	9602	Human milk
11100000	Milk, NFS	1004	Milk, reducr
11111000	Milk, whole	1002	Milk, whole
11112110	Milk, reduced fat (2%)	1004	Milk, reducr
11112210	Milk, low fat (1%)	1006	Milk, lowfat
11113000	Milk, fat free (skim)	1008	Milk, nonfat
11114300	Milk, lactose free, low fat (1%)	1006	Milk, lowfat

* [Ingredient Nutrient Values.xls](https://www.ars.usda.gov/ARSUserFiles/80400530/apps/2019-2020%20FNDDS%20At%20A%20Glance%20-%20Ingredient%20Nutrient%20Values.xlsx)

<https://www.ars.usda.gov/ARSUserFiles/80400530/apps/2019-2020%20FNDDS%20At%20A%20Glance%20-%20Ingredient%20Nutrient%20Values.xlsx>

Import the sheet **Ingredient Nutrient Values**

The screenshot shows the Power BI Navigator window. On the left, under 'Display Options', the 'Ingredient Nutrient Values' sheet is selected with a blue border and a green checkmark. On the right, a preview of the sheet is shown, titled 'Ingredient code' and 'Ingredient description'.

Ingredient code	Ingredient description
1001	Butter, stick, salted
1001	Butter, stick, salted
1001	Butter, stick, salted

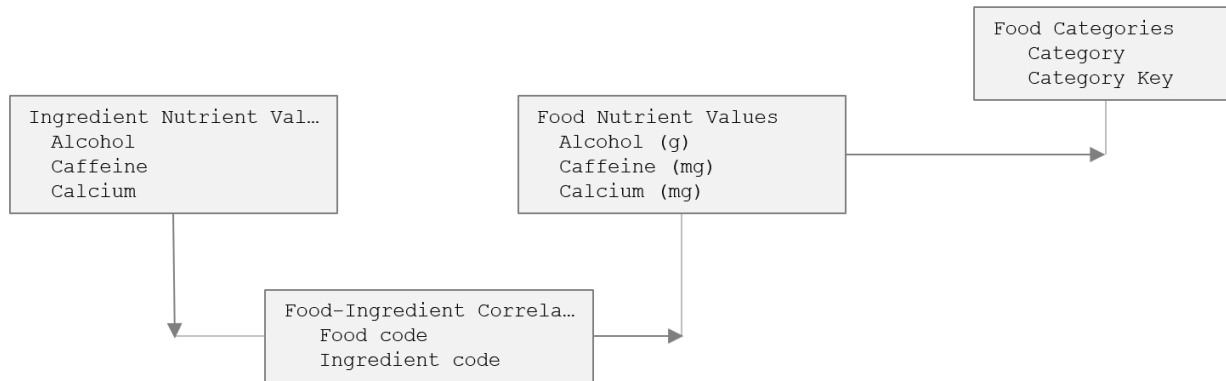
- * Note: If for some reason you have trouble with the Web import option, alternatively you may download each file and then import it to your project.
3. Open Power Query using the Transform Data button.
 4. In the FNDDS_Nutrient_Values table, perform the following transformations:
 - * Change column names:
 - Main food description → Food
 - WWEIA Category description → Category
 - * Delete columns:
 - WWEIA Category number
 - All columns with headers using format *N:N (g)*, **4:0(g)** through **22:6 n-3 (g)**
 5. In the Ingredient table:
 - * Apply “Use first row as headers”
 - * Change column name
 - Ingredient description → Ingredient
 - * Keep columns (ctrl+click to select, then use “Remove Other Columns”):
 - Ingredient code
 - Ingredient
 - Nutrient description
 - Nutrient value
 6. What do you notice about columns versus rows between the two tables?
 - * Pivot the **Nutrient description** column, setting “Values Column” to **Nutrient value**
 7. So far, your two existing tables don’t have a way to relate to each other. You’ll need to import some additional data to draw this relationship:
 - * Import using Get Data > Web:
[FNDDS Ingredients.xlsx](https://www.ars.usda.gov/ARUserFiles/80400530/apps/2019-2020%20FNDDS%20At%20A%20Glance%20-%20FNDDS%20Ingredients.xlsx)
<https://www.ars.usda.gov/ARUserFiles/80400530/apps/2019-2020%20FNDDS%20At%20A%20Glance%20-%20FNDDS%20Ingredients.xlsx>
 8. Transform the new data:

- * Apply “Use first row as headers”
 - * Select Food code and Ingredient code, and remove all other columns
9. Break out Category from the main foods table to a separate related table:
- * Right-click query FNDDS Nutrient Values and select Duplicate. Rename as Food Categories.
 - * Select Category Key and Category, and remove all other columns
 - * Right-click column header for Category Key, and select Remove Duplicates
10. Go back to FNDDS table and delete Category column
11. Do a little cleanup to rename your queries more clearly:
- * FNDDS Nutrient Values → Food Nutrient Values
 - * FNDDS Ingredients → Food-Ingredient Correlation
12. Save project as NutritionData_Step01.pbix

Modeling the Data

13. Click Close & Apply to return to Power BI, then go to the Model view. Rearrange tables to fit the page (or adjust zoom level).
14. Note that Power BI has detected two relationships.
- * What key is used for each relationship?
 - * What’s the cardinality? How can you tell?
15. Create a relationship between Food Nutrient Values and Food-Ingredient Correlation
- * Locate the Food code field in each card
 - * Drag Food code from one card onto the same field in the other to create the relationship

16. Adjust the arrangement of your tables to look something like this:



Applying the Data to a Scenario

So far, none of the tables in this data model include dates – it’s just descriptive information. It’s time to introduce a scenario where we wish to apply this data.

You are working with a pediatric nutritionist’s office to help them create a system for creating client data reports. For this first report, the client is Goldilocks, a 10-year-old girl who has a history of strong food aversion. When Goldilocks’s parents brought her in for her first consultation, she would only eat porridge and a few other limited items. The nutritionist worked with her family over a period of months to introduce new foods into Goldilocks’s diet. You have been provided with a set of “food journal” files that track these changes.

17. **Food Journal Data** is available as a zip file in the course material. Please download and unzip, and add the folder to your homework directory for week-11.
18. Use **Get Data > Import from folder** to import Food Journal Data (should contain two files). Then use **Combine & Transform Data** to combine the .csv files.
19. In Power Query, take a look at the columns in this new table:
 - * We will not use Source.Name in this exercise, so this column can be deleted. (Consider: When do you think you might use this information?)
 - * Also note the “Transform File” helper queries that were automatically added.
20. Click **Close & Apply**, and return to the Model View. Expand the table **Food Journal Data** in the Data panel at the far right.

- * Review the Date Hierarchy created by auto date/time (if there is one).
 - * Revisit your schema. What should be the main fact table?
21. Click Transform Data to head back to Power Query, and create a quick date table using M language.
- * Right-click in the blank area of the queries panel on the left, and choose New Query > Blank Query. Change the query name to Date.
 - * In the query box at the top center, type in the below expression, then hit enter:

```
= List.Dates(#date(2022, 11, 25), 160, #duration(1, 0, 0, 0))
```
 - * In the ribbon, click To Table to convert this list of dates to a table.
 - * Rename the column to Date, and change its data type to date.
22. With the Date column selected, go to the Add Column tab of the ribbon. Use the Date drop-down to add columns for Year, Month Name, Month Number, Day of Week Number, and Day Name. (Use the appropriate options to add each column, then rename columns as needed.)
23. Click Close & Apply, and return to the Model View.
24. Add a relationship between Food Journal Data and Date tables using the Date field.
25. Update column “sort by” for Month Name and Day of Week so that they are sorted by number.
26. Add one more supporting table: **Food Journal Administration.csv** (from course material downloads).
27. In the Model View, draw a relationship from this new table to the Date table.
28. Save project as NutritionData_Step02.pbix

Enhancing the Model with DAX Calculations

29. Switch over to Table View and expand the Food Journal Data table.

30. Create the following new measures Food Journal Data:

- * **New measure #1:**

```
Unique Foods Eaten = DISTINCTCOUNT('Food Journal Data'[Foods])
```

- * **New measure #2 (rolling total):**

```
Total Foods Tried =
CALCULATE(
    DISTINCTCOUNT('Food Journal Data'[Foods]),
    FILTER(
        ALL('Food Journal Data'),
        'Food Journal Data'[Date] <=
        MAX('Food Journal Data'[Date])
    )
)
```

31. Add a calculated column to Food Journal Data

- * Currently, the Meal column describes the meal when a food was consumed by its text name.
- * Create a column using the logical function SWITCH() to convert the names of each mealtime to a number value, to provide an alternative option for sorting/labeling.
- * Use the Column Tools > New Column to add a column with the following formula:

```
Meal Number = SWITCH(
    'Food Journal Data'[Meal],
    "Breakfast", 1,
    "Lunch", 2,
    "Dinner", 3, "Snack", 4)
```

- * **BONUS:** What happens if you head to the Model View and try to adjust the “sort by” order for the Meal column to use Meal Number? How could you create Meal Number differently so that it could be used for “sort by”?

32. Add Row-Level Security: In the ribbon Home tab, start by going to Manage Roles.

- * Create a new role called Patient, and select the Food Journal Administration table
- * Use the ellipses icon for Food Journal Administration to select Add Filter > [Role]
- * Add appropriate logic to limit the Patient view to only patient (not office or administrator) data.
- * Click Save when you are done.

33. Back in the Data View, look at the Food Journal Administration table.

- * If you scroll down, you'll see multiple different roles listed in the Role column (may have to scroll down far).

34. Now click View As in the ribbon, and check the box for Patient.

- * You'll see the banner "Now viewing as: Patient"
- * If you scroll through the Food Journal Administration table again, this time you'll only see the rows with a Role value of Patient.

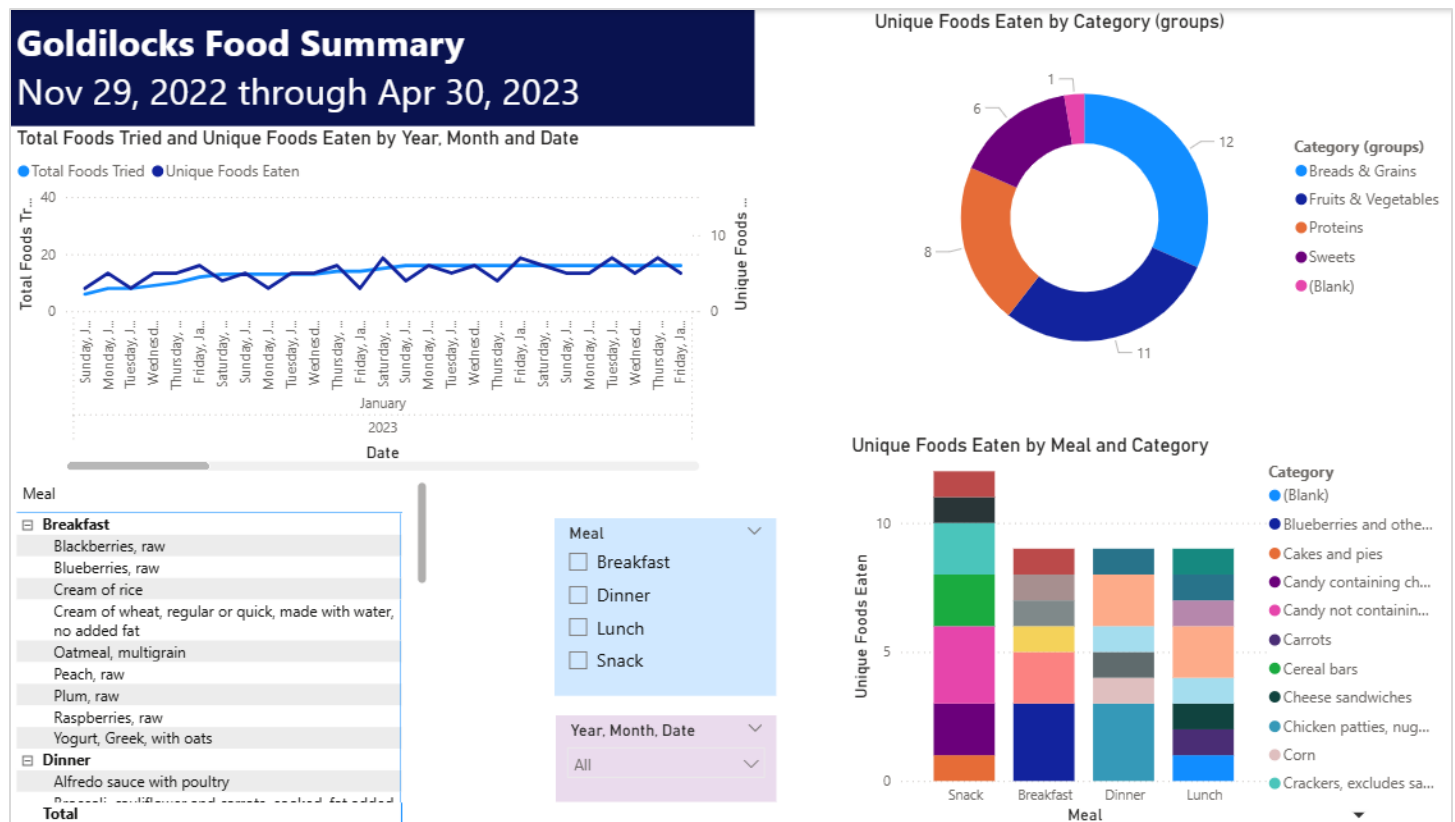
35. Save project as NutritionData_Step03.pbix

Exercise 3.B Visualizing Nutrition Data Results

Adding Visualizations

1. Create the following visualization types for the report, using the values / fields provided below.

* Also feel free to customize formatting for each visual. You are welcome to get creative!



2. Text box:

* Goldilocks Food Summary, Nov 29 2022 through Apr 30 2023

3. Line chart:

- * X-axis: Date > Date Hierarchy
- * Y-axis: Food Journal Data > Total Foods Tried
- * Secondary y-axis: Food Journal Data > Unique Foods Eaten

4. Donut chart:

- * Legend: Food Categories > Category
- * Values: Food Journal Data > Unique Foods Eaten

5. Stacked column chart:

- * X-axis: Food Journal Data > Meal
- * Y-axis: Food Journal Data > Unique Foods Eaten
- * Legend: Food Categories > Category
- * Also try changing this to a 100% stacked column chart. When/why might you use either version?

6. Matrix:

- * Rows: Food Journal Data > Meal
- * Rows: Food Journal Data > Foods
- * Columns: <none>

7. Slicer:

- * Field: Food Journal Data > Meal

Grouping Data

8. Select your donut chart, Unique Foods Eaten by Category:
9. Under Legend, click the down-chevron by Category and select **New Group**.
10. Use the pick-list of Category values to create four groups, renaming each as provided below (note that you can use ctrl+click to multi-select items):
 - * **Fruits & Vegetables**
Blueberries and other berries
Carrots
Corn

Other fruits and fruit salads
Other vegetables and combinations
Peaches and nectarines

* **Proteins**

Chicken patties, nuggets and tenders
Dips, gravies, other sauces
Other Mexican mixed dishes
Yogurt, Greek

* **Sweets**

Cakes and pies
Candy containing chocolate
Candy not containing chocolate

* **Breads & Grains**

Cereal bars
Cheese sandwiches
Crackers, excludes saltines
Grits and other cooked cereals
Oatmeal
Pasta mixed dishes, excludes macaroni and cheese
Pasta, noodles, cooked grains
Pizza
Pretzels/snack mix

Creating a Drill-through Page

11. Add a new report page using the + tab at the bottom of the canvas

- * Rename your new page Food Groups
- * While you're at it, rename your first page Summary

12. Add a table visualization to the page using two values for Columns:

- * Food Categories > Category
- * Food Journal Data > Foods

13. For your table, into the field marked Add drill-through fields here, drag:

- * Food Categories > Food Groups

- * Note that Power BI will automatically add a back arrow icon/link to the top corner of your page. Reposition your visualization as needed to make the back button clearly visible.

14. Try it! On the Summary page, right click a segment of the donut chart and select Drill through > Food Groups

15. Save project as NutritionData_Step04.pbix

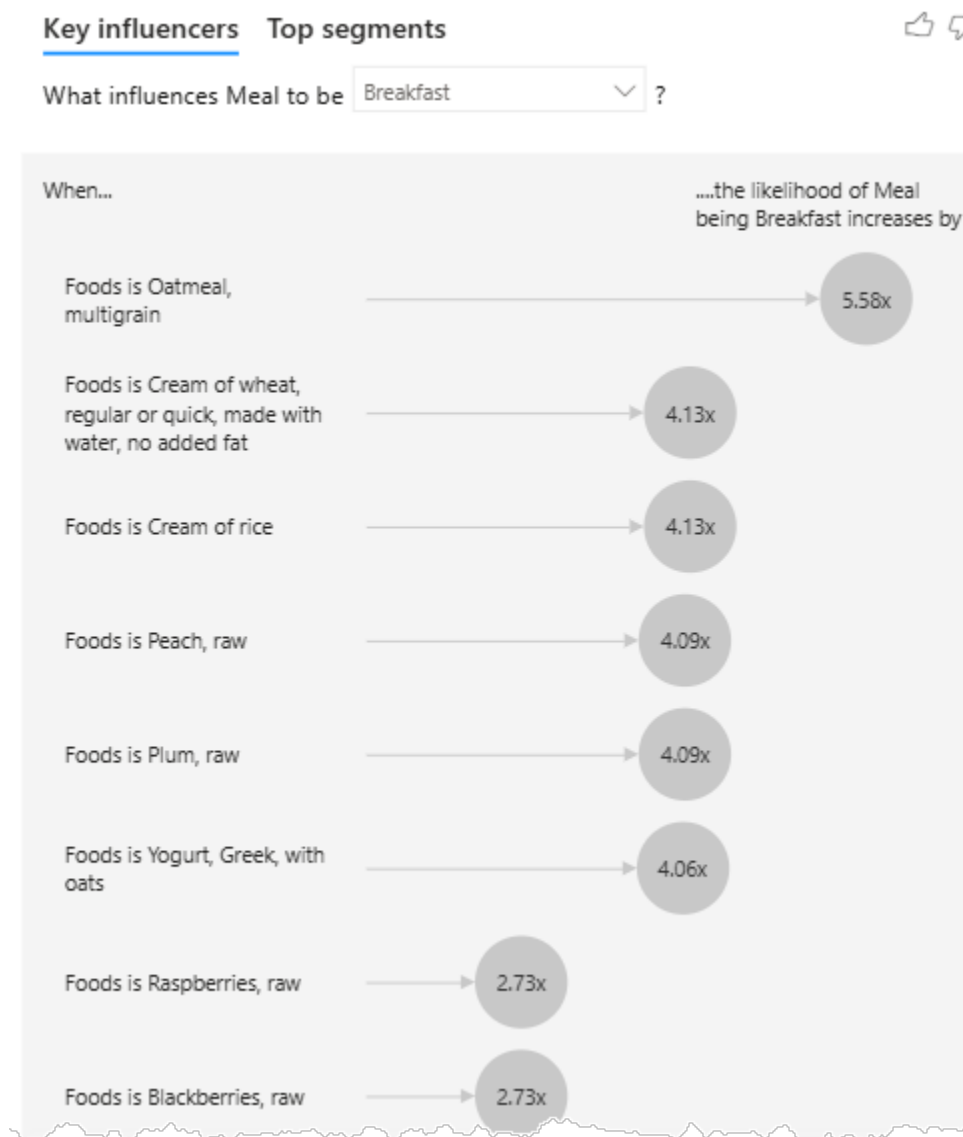
Module 4

Advanced Analytics with Power BI

Exercise 4.A Using AI-enabled Visualizations

Add a Key Influencers Visualization

1. Add a new page to the nutrition data report, and name it Meal Breakdown.
2. Add a Key Influencers visualization using the  icon.
3. In the data wells, for **Analyze** add the Meal field. For **Explain by** add Foods.
4. Explore how the visualization changes when you select different meals from the “What influences” drop-down at the top:

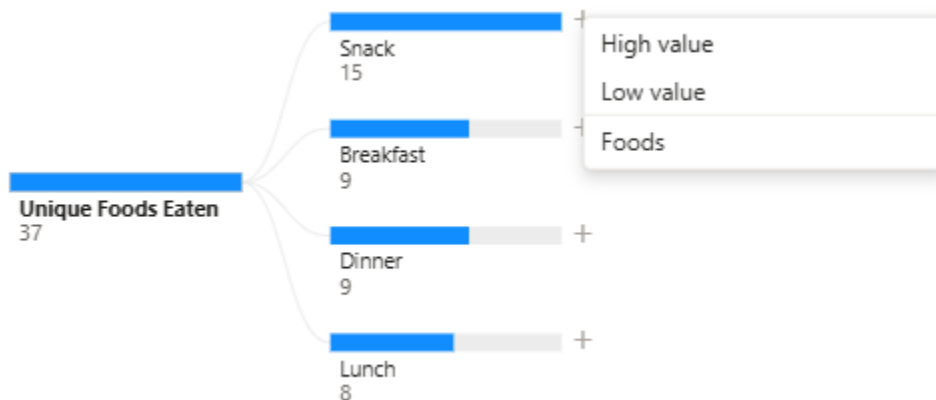


Add a Decomposition Tree

5. Also on the Meal Breakdown tab of the report, add a Decomposition Tree visualization using the  icon.
6. In the data wells:
 - * For **Analyze** add Unique Foods Eaten
 - * For **Explain by** add Meal and also Foods
7. Explore how you can expand the decomposition tree by clicking the plus sign next to Unique Foods Eaten, then clicking on Meal:



8. Then try expanding a particular meal by Foods:



9. Review the breakdown of different foods eaten by meal. Experiment with using the X at the top to remove a level and then expanding out by a different meal name. Also try sorting by High value versus Low value.
10. Save project as NutritionData_Step05.pbix

Module 5

Sharing Power BI Reports

Exercise 5.A Sharing the Nutrition Data Report

Publishing the Report

1. To publish your Nutrition Data report:
 - * Export as a PDF file and save to week-11
 - * Publish to Power BI service using My Workspace
2. From the pop-up confirmation that your report has been published to Power BI service, click to view your report online (app.powerbi.com)
 - * Use the Share button to email the report to your instructor. (You will need to activate your free trial of Power BI premium.)
 - * Using the ... icon from the toolbar above the report, select **Pin to a dashboard** and pin the report page to a **new** dashboard named **Goldilocks Dashboard**.
 - * Hover over individual visualizations in the report, and choose a few specific visualizations to pin to the dashboard as well.
3. On the left rail, navigate to My Workspace, and spend a little time exploring:
 - * In the workspace, find your “nutrition data” semantic model.
 - * Use the ... icon to select **Get quick insights**. This will take a few minutes, so move on to the next step for now.
 - * Use the ... icon to open **Settings**, and explore the options available for your semantic model.
4. Navigate back to My Workspace. Find your nutrition data model again, and this time from the ... menu, go to View Insights.
 - * Review the machine-learning generated insights for your dataset.
 - * Choose a few visuals to pin to Goldilocks Dashboard.
5. Again navigate back to My Workspace. Click into Goldilocks Dashboard.
 - * Review the dashboard. How can you edit it? Arrange the dashboard how you like.

- * Check the mobile layout for your dashboard. Are there any edits you can make here to make the dashboard more readable in this compact format?
- * When you are done making changes, use the **Share** option to send the dashboard to your instructor's email address, checking all boxes in the sharing options.

End of Week 11 Lab Workbook